

Phase 3 - Project Design Phase

FitBuddy follows a modular architecture connecting the frontend, FastAPI backend, Gemini AI services, and SQLite database.

System Architecture

- Frontend: HTML and Jinja2 templates.
- Backend: FastAPI for routes, form processing, and application logic.
- AI Layer: Google Gemini models for workout plans, updates, and nutrition tips.
- Database: SQLite with SQLAlchemy for persistent storage.

Frontend Design

- index.html - collects user information.
- result.html - displays the workout plan, nutrition tip, and feedback form.
- all_users.html - displays users and their original/updated plans.

AI Integration Design

- Gemini Pro is used for structured workout generation and feedback-based updates.
- Gemini Flash is used for fast nutrition or recovery tips.
- AI responses are returned to the frontend and displayed in a readable format.

System Flow

- User Input → FastAPI → Gemini AI → Workout Plan + Nutrition Tip → Database → User Feedback → Updated Plan.

Phase	Focus
3	Project Design Phase